



# Jyoti Prakash

Bachelor of Technology  
Electronics and Communication Engineering  
National Institute of Technology, Patna

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[github.com/jyotiprakash2033](https://github.com/jyotiprakash2033)

[leetcode.com/u/Jyotiprakash2033](https://leetcode.com/u/Jyotiprakash2033)

## PROFESSIONAL SUMMARY

Software Engineer with hands-on experience in mobile application development, backend services, and AI applications. Built a React Native EdTech platform, developed backend features using Node.js, Express.js, PostgreSQL, and AWS S3, and implemented a Retrieval-Augmented Generation (RAG) assistant and a custom 64-bit RISC-V processor. Interested in scalable backend systems and AI-driven software development.

## EDUCATION

| Degree/Certificate | Institute/Board                            | CGPA/Percentage | Year           |
|--------------------|--------------------------------------------|-----------------|----------------|
| B.Tech, ECE        | National Institute of Technology, Patna    | 8.11 CGPA       | 2027(Expected) |
| Senior Secondary   | Teresa International Academy, Patna / CBSE | 85.8%           | 2022           |
| Secondary          | Shivam School, Bihta / CBSE                | 92.8%           | 2020           |

## EXPERIENCE

- **Atomverse** [[Website](#)] [[Certificate](#)] July 2024 – January 2025  
Patna, India  
VR Developer Intern
  - Developed Unity3D and C# VR learning modules covering 4 interaction patterns: spatial navigation, quizzes, object interaction, and dynamic UI.
  - Reduced polygon count by 30% and improved frame rates by 25% across standalone and low-end VR headsets through asset and scene optimization.
  - Engineered 4 reusable C# component groups for teleportation, interaction, quiz flows, and UI overlays, enabling faster learning-module updates without core code changes.
  - Standardized cross-device input and rendering compatibility layers to improve reliability across multiple VR hardware targets.

## PROJECTS

- **Prayas IAB – Scalable Content Platform (Freelance)**  
*Technologies: React Native, Expo, Redux Toolkit, Node.js, Express.js, PostgreSQL, Prisma, AWS S3*
  - Developed a React Native EdTech platform with a six-level academic hierarchy supporting categories, classes, modules, subjects, chapters, quizzes, and video lessons.
  - Designed normalized PostgreSQL schemas using Prisma ORM and developed scalable RESTful APIs with Node.js and Express.js for authentication, content delivery, analytics, and progress tracking.
  - Implemented Redux Toolkit state management and role-based access control (RBAC) for three role-specific experiences: students, admins, and content managers.
  - Delivered a timed quiz engine with auto-scoring and AWS S3 cloud storage for protected solution-video streaming.
- **AI Knowledge Assistant – RAG-Powered Q&A System** [🔗]  
*Technologies: Python, FastAPI, React.js, Gemini API, Ollama, Qdrant, Hugging Face, Crawl4AI*
  - Built a Retrieval-Augmented Generation (RAG) assistant using FastAPI, Gemini API, Ollama, Hugging Face embeddings, and Qdrant for document and web-based question answering.
  - Implemented a 6-stage retrieval pipeline: ingestion, chunking, embedding generation, Qdrant indexing, semantic search, and response generation.
  - Integrated Crawl4AI to extract, process, and index external web content into the vector database for expanded knowledge coverage.
  - Developed and documented REST APIs for document upload, website crawling, retrieval, and conversational chat, integrated with a React frontend.
- **64-bit RISC-V Processor with ML Accelerator** [🔗]  
*Custom RV64 processor with memory-mapped ML acceleration for inference and training.*
  - **Technologies:** RISC-V, Verilog RTL, Icarus Verilog, GTKWave, Python
  - Implemented RV64I/M/B processor with decode, ALU, register file, control, and memory interface
  - Designed memory-mapped ML accelerator with SIMD int8 (8 MACs/cycle), dot-product/GEMM engines, and DMA; no ISA changes
  - Verified with 9,900+ differential checks against independently generated RISC-V reference models
  - Ran regression/MLP inference and fixed-point gradient training in integer arithmetic, reducing 784-feature workload from 3,100 to 48 instructions

## CODING PROFILES

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◦ **DSA Practice:** 350+ problems solved across [LeetCode](#) (200+), [GeeksforGeeks](#) (80+), and [CodeChef](#) (40+).

## TECHNICAL SKILLS

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- **Programming Languages:** Python, JavaScript , C++, C, Verilog
- **Frontend & Mobile:** React.js, React Native, Expo, Redux Toolkit, HTML, CSS
- **Backend & APIs:** Node.js, Express.js, RESTful APIs, JWT Authentication, Prisma
- **Databases:** PostgreSQL, MongoDB,
- **AI / LLM:** Retrieval-Augmented Generation (RAG), LLM Integration, Gemini API, Ollama, Hugging Face Embeddings, Semantic Search, Prompt Engineering
- **Cloud & Tools:** Git, GitHub, API Testing (e.g. Postman), Unity3D, Blender, GTKWave, Icarus Verilog

## AWARDS & ACHIEVEMENTS

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- **MSME Idea Hackathon 4.0 (2024):** Selected by the Ministry of MSME, Government of India, as a team member for an innovation project under the national idea hackathon.

## CERTIFICATIONS

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|--------------------------------------------------------------------------------------|------|
| ◦ NPTEL (IIT Madras), <a href="#">The Joy of Computing using Python</a> – Elite, 85% | 2025 |
| ◦ NPTEL (IIT Kharagpur), <a href="#">Cloud Computing</a> – Elite, 82%                | 2026 |
| ◦ NPTEL (IIT Guwahati), <a href="#">Digital Design with Verilog</a> – Elite, 73%     | 2026 |

## LEADERSHIP & EXTRACURRICULARS

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- **Team Lead, Astomverse Community** – Led a team of 10+ in building interactive VR simulations at NIT Patna.
- **Member, DFC Club** – NIT Patna 2024 – Present